

Java Full Stack Syllabus

Course Duration: 120 hrs.

HTML

S. No	Content
1	HTML • Introduction of HTML • Tags in HTML • Basic Structure • Heading tags • Paragraph tag <p> • Creating a simple website
2	HTML Formatting • What is formatting • creating Bold text, Italic text, Strong text. • Inserting new data • Deleting old data • highlighting text • Creating address field in website • Location or site field • Writing a quotes in website • Creating Blog website using Formatting

3	HTML Attribute • Style • Class • Id • Span • Width and Height • Source • Alternative • Relative • href attribute
4	HTML Link • What is HTML link • How to add website link • How to connect other pages in HTML page • External Reference • Creating Hyper Link • Attributed used in Hyper link • How to create image link

5	HTML Media • Adding Images • Adding animated images like Gif • Adding Audio • Adding video
6	HTML List 1. What is List 2. How to create list in web page Types: 3. Ordered List 4. Unordered List 5. Description List 6. Horizontal List (Nav Bar) • Creating a Navigational bar in our web site.
7	HTML Semantic Tags • What is semantic tags • Semantic tags in HTML5 • Block Elements • Inline Elements

8	HTML Layout • Header • Main content • Side content • Session □ Articles • Nav bar • Footer
9	HTML Responsive website • What is responsive web site • How we can create Responsive website. • Creating Responsive text • Creating Responsive image • Attributes used in Responsive website
10	HTML Form • What is Form • Tags used to create form • Attributes used in form • Input tags and its types • Creating radio button • Creating check box • Creating dropdown boxes • Creating date calendar • Creating an Enquiry form web site.

11.	HTML Table • What is Tables • Why we use tables in website • Creating a table • Adding rows and columns
12	Exercise and Practice

CSS

1	CSS • Introduction of CSS • Selector, Declaration • Property, Value • Grouping selector
2	CSS Types • Inline • Internal • External
3	CSS Color • RGB Values • HSL Values • HEX Value • RGBA • HSLA

4	CSS Background • How to decorate background in web page • Adding Background Image • Background-repeat property • Changing Background size • Transparency of background using opacity • Adding Gradient color • Box-sizing
5	CSS Margin, • What s margin • How we can use it • Using top, right, left and bottom property • Values assigning for margins CSS Padding • What s margin • How we can use it • Using top, right, left and bottom property • Values assigning for margins CSS Border • What s margin • How we can use it • Using top, right, left and bottom property • Values assigning for margins

6	CSS Shadow Effect What is shadow effect Types: 1. Text Shadow 2. Box Shadow 3. Multi Shadow
7	CSS Text • Text color • Text decoration • Text transform • Text indent • word spacing • Letter spacing
8	CSS Font • Generic Family • Font-family • Font style • Font size • Font weight • Font variant

9	CSS Table • Border property • Border-collapse property CSS Outline • What is outline • Outline-style properties • Outline-color • Outline-width CSS Dropdown • What is dropdown • Dialogue box • Dropdown List
10	CSS Layout • Creating layout using css properties • Grid Layout • Grid layout properties • Flex Layout • Flex layout properties
11	CSS Transforms & Transition • What is transforms • What is transition • How we use it • Using properties

12	CSS Animation • What is Animation • How animation works • @Keyframe keyword • Example for Animations in CSS
13	Exercise & Projects.

Bootstrap

S. No	Content
1	Bootstrap • Introduction to Bootstrap • Features • Using CDN • Downloading bootstrap as module • Using bootstrap to create a website • Adding Icons using bootstrap

Java Script

S. No	Content
2	Java Script • Introduction to JavaScript • Client and server side language • Feature • Hello world program • Types a. Internal b. External
3	Variable • What is variable • Declaration of a variable Types: 1. Local 2. Global Data Types • What is Datatype Types: • Primitive • Non-Primitive

4	Operators • Arithmetic • Assignment • Relational • Misc • Logical • Conditional
5	Decision Making • Introduction • If Statements a. If – else b. Nested If c. If else ladder • Switch Case.
6	Flow Control • Introduction • 3 major steps in loop • Types: 1. For Loop 2. While loop 3. Do-While loop.

7	Function Introduction to functions Declaration of a function Types: • With argument with Return • Without argument with Return • With argument without Return • Without Argument without Return
8	Event Handling Introduction to Events How to create an Event Example: 1. OnClick 2. OnChange 3. OnSubmit 4. MouseOver 5. Confirmation
9	JS Object • What is Objects • Math Object • Object Creation

10	JS Time Format • How we can use time and date in website • Date() Constructor • Finding date, hours, minutes and seconds • Creating a running clock
11	Exercise and Projects

Core Java

S. No	Content
1	Introduction of Java • What is Java • Who Developed Java • How Java runs • Features of Java • Applications of Java • Types of Java
2	Fundamentals of Java • Main function in java □ Hello World Program • Variables • Naming rule for variables • Variable Declaration

3.	Datatypes What is Datatypes Declaring a datatype Types: 1. Primitive 2. Non- primitive
4.	Operators • Arithmetic • Assignment • Relational • Increment / Decrement • Logical • Conditional
5.	Decision Making If Statements a. If – else b. Nested If If else ladder Switch Case.
6	Flow Control • For Loop • While loop • Do-While loop.

7	Methods • Introduction to methods • How to create a method • Uses of methods • Types of Methods
6	Array • Introduction to array • How to declare a array • Getting user input for array • Types: 1. Single Dimensional 2. Multi-Dimensional
7	String • Introduction to String • Declaration of String • String methods • String Pool String Buffer • Introduction to String Buffer • Declaration of String Buffer • String Buffer methods

	String Builder • Introduction to String Builder • Declaration of String Builder • String Builder methods • Difference between String, String Buffer & String Builder
8	Object Oriented Programming - OOPs • Introduction of OOPs • Concepts or types of OOPs • Access Modifier • Class & Objects
9	Inheritance Introduction to inheritance Using extends keyword Types: 1. Single 2. Multi-level 3. Hierarchy
10	Abstraction • Introduction to abstraction • Using abstract keyword • Declaration of abstract method • Defining abstract method

11	Interface • Introduction to interface • Creating an interface using interface keyword • Declaring a variable and method • Multiple - inheritance
12	Encapsulation • Introduction to Encapsulation • Using private keyword • Setter & getter Method
13	Polymorphism • Introduction to polymorphism • Method Overloading • Constructor Overloading I. Introduction to constructor II. Creating a constructor III. Types: • Default Constructor • Parameterized Constructor • Method Overriding

14	Exception Handling • What is Exception • How we can handle the exception • Keywords: 1. Try Catch 2. Throw 3. Throws 4. Finally
15	Exercise & Projects.

Advance Java & J2EE

S. No	Content
1.	MultiThreading • What is Thread • What is Multi-Tasking • How Multithreading works • Life Cycle of thread • Creating a thread: 1. Using Thread class 2. Using Runnable interface 3. What is Daemon Thread • How to create daemon thread • Synchronized Keyword

	• Converting a normal method as thread • Creating more than one thread • What is Garbage Collection
2	Collection Framework: • Introduction to collection • Use of Collections • Collection Object • Collection classes and interfaces List Interface: 1. Introduction to List 2. How it stores the data in memory 3. Types: ○ ArrayList ○ LinkedList ○ Stack ○ Vector • Set Interface 1. Introduction to Set 2. How it stores the data in memory 3. Types: ○ HashSet ○ TreeSet • Queue Interface 1. Introduction to Set

	2. How it stores the data in memory 3. Types: ○ Deque ○ PriorityQueue
3	Map Interface 1. Introduction to Set 2. How it stores the data in memory 3. Types: ○ HashMap ○ TreeMap ○ LinkedHashMap ○ NavigableMap
4	I/O Stream • Introduction to I/O Stream • What Input Stream • What Output Stream • Creating a file • Reading and Writing in a file • Closing and Deleting a file
5	Serialization & Deserialization • Introduction • Encrypting the data • Decrypting the data

6	Networking • What is networking • Network Topologies • What is Socket • Use of socket class • One way communication • Two way communication • Finding Inet Address
6	Java Swing • Swing introduction • Swing Classes • Creating a frame • GUI Application • Creating form, input, radio button, dropdown • Adding images in frame and button • What is Event Handling • How Events work
7.	Layout Manager Introduction to Layout Creating a frame with layout Types of Layout: 1. Grid Layout 2. Box Layout 3. Card Layout 4. Flex Layout

8	JDBC • Java DataBase Connectivity Introsuction • MySQL Introduction • MySQL Administraction • Using SQL Queries. • Creating table and adding columns • Using Constrains in SQL • Inserting data, updating and deleting the tables. • Connecting Java to MySQL • CRUD – Create Read Update Delete Operation
9	Exercise & Projects.

Spring Boot

1	Spring Boot • What is Spring Boot • Features of Spring Boot • Creating a Spring Boot project using Spring Initializr • Software Setup and Configuration (Eclipse) • How to Run the Project
2	Spring Boot Core • What is Spring Core • IoC Container • BeanFactory Container • ApplicationContext Container

	• Spring Boot Architecture • Layers in Spring Boot
3	Spring Boot Annotations • What is Annotation • Annotation List • Controller Annotation • Request Handling (HTTP Request)
4	Spring Boot with REST API • What is REST API • How it works • REST Controller • Example Program
5	Spring Boot with Database and Data JPA • What is Spring JDBC • JDBC Connecting Pool • Driver Connection
6	Spring Boot with Micro services • What is MicroServices • Architecture of MicroServices • Principle of MicroServices
7	Exercise & Practice

SQL

1	Database • What is Database or DBMS • What is RDBMS • Types of SQL Databases • Database Normalization • Types of Normalization • Installing MySQL and Configurations
2	Java MySQL Database Access • Database Handling with MySQL • Create Database Connection • SQL Constrains • SQL Syntax
3	DDL Operations • What is DDL command • Creating database • Creating table • Dropping Table • Dropping Database • Altering the table • Using Truncate

4	DML Operations • What is DML • Selecting the row or column • Inserting data into table • Updating the data in table • Deleting data or row, columns
5	DCL Operations • What is DCL • Grant • Revoke
6	TCL Operations • What is TCL • Commit • Rollback • Savepoint
7.	SQL DataTypes SQL Operators • Arithmetic • Relational • Logical •

8	SQL Expression • Boolean • Numeric • Date SQL Cloning SQL View SQL Clause
9	SQL Joins • What is Joins • Inner Join • Outer Join 1. Left Join 2. Right Join . Full Join • Cross Join • Self Join • Union
9	Exercise & Practice
	Complimentary • 10+ Projects • Maven • Postman • GitHub • Stack Overflow